

วิธีการรันและผลลัพธ์

1.เปิด Docker containers ทั้งหมด

- docker compose up -d

```
MINGW64~/OneDrive/Documents/GitHub/DAD56002/HW (main)
$ docker compose up -d
[+] up 12/12
✓ Network hw_default Created 0.0s
✓ Container hive-metastore-postgresql Started 0.73s
✓ Container nodemanager Started 0.85s
✓ Container datanode Started 0.65s
✓ Container historyserver Started 0.85s
✓ Container resourcemanager Started 0.85s
✓ Container spark-master Started 0.96s
✓ Container namenode Started 0.96s
✓ Container hive-metastore Started 0.96s
✓ Container spark-worker Started 1.25s
✓ Container hive-server Started 1.25s
✓ Container hue Started 1.55s

What's next:
  Filter, search, and stream logs from all your Compose services
  in one place with Docker Desktop's Logs view. docker-desktop://dashboard/logs?appId=hw

MINGW64~/OneDrive/Documents/GitHub/DAD56002/HW (main)
$
```

2.รัน MapReduce Job

- MSYS_NO_PATHCONV=1 ./execute.sh

3.ดูผลลัพธ์ MapReduce

- MSYS_NO_PATHCONV=1 docker exec -it namenode hdfs dfs -cat /user/root/output/part-00000

MapReduce Output

```
boomn@DESKTOP-IU0AVN0 MINGW64 ~/OneDrive/Documents/GitHub/DAD56002/HW (main)
$ MSYS_NO_PATHCONV=1 docker exec -it namenode hdfs dfs -cat /user/root/output/part-00000
2026-06-02 11:13:00,649 INFO Sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteHostTrusted = false
1      163214.29
2      103600.00
3       59300.00
4      30571.43
5       20888.89

What's next:
  Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug namenode
  Learn more at https://docs.docker.com/go/debug-cli/

boomn@DESKTOP-IU0AVN0 MINGW64 ~/OneDrive/Documents/GitHub/DAD56002/HW (main)
$
```

ขั้นที่ 4: รัน Spark Job

- MSYS_NO_PATHCONV=1 ./spark.sh

Spark RDD Output

```
bin\windows\MSYS_NO_PATHCONV=1 MINGW64 ~/OneDrive/Documents/gitHub/DAD66002/hw (main)
$ MSYS_NO_PATHCONV=1 ./spark.sh
Using Spark's default log4j profile: org/apache/spark/log4j-defaults.properties
26/06/02 11:15:00 INFO SparkContext: Running Spark version 3.2.1
26/06/02 11:15:00 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
26/06/02 11:15:00 INFO ResourceUtils: =====
26/06/02 11:15:00 INFO ResourceUtils: No custom resources configured for spark.driver.
26/06/02 11:15:00 INFO ResourceUtils: =====
26/06/02 11:15:00 INFO SparkContext: Submitted application: Assignment2.AverageIncome_RDD
26/06/02 11:15:00 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, amount: 1, script: , vendor: , memory -> name: memory, amount: 1024, script: , vendor: , of
Heap -> name: offHeap, amount: 0, script: , vendor: ), task resources: Map(cpus -> name: cpus, amount: 1.0)
26/06/02 11:15:00 INFO ResourceProfile: Limiting resource is cpu
26/06/02 11:15:00 INFO ResourceProfileManager: Added ResourceProfile id: 0
26/06/02 11:15:00 INFO SecurityManager: Changing view acls to: root
26/06/02 11:15:00 INFO SecurityManager: Changing modify acls to: root
26/06/02 11:15:00 INFO SecurityManager: Changing view acls groups to:
26/06/02 11:15:00 INFO SecurityManager: Changing modify acls groups to:
26/06/02 11:15:00 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view permissions: Set(root); groups with view permissions: Set(); users with modify permissions:
Set(root); groups with modify permissions: Set()
26/06/02 11:15:01 INFO Utils: Successfully started service 'sparkDriver' on port 33147.
26/06/02 11:15:01 INFO SparkEnv: Registering MapOutputTracker
26/06/02 11:15:01 INFO SparkEnv: Registering BlockManagerMaster
26/06/02 11:15:01 INFO BlockManagerMasterEndpoint: Using org.apache.spark.storage.DefaultTopologyMapper for getting topology information
26/06/02 11:15:01 INFO BlockManagerMasterEndpoint: BlockManagerMasterEndpoint up
26/06/02 11:15:01 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
26/06/02 11:15:01 INFO BlockManager: Created local directory at /tmp/blockmgr-ec06a0d-8af2-4bd2-b4f8-a24850e73473
26/06/02 11:15:01 INFO MemoryStore: MemoryStore started with capacity 366.3 MiB
26/06/02 11:15:01 INFO SparkEnv: Registering OutputCommitCoordinator
26/06/02 11:15:01 INFO Utils: Successfully started service 'sparkui' on port 4040.
26/06/02 11:15:01 INFO SparkUI: Bound SparkUI to 0.0.0.0, and started at http://7362f7467f92:4040
26/06/02 11:15:01 INFO Executor: Starting executor ID driver on host 7362f7467f92
26/06/02 11:15:01 INFO Utils: Successfully started service 'org.apache.spark.network.netty.NettyBlockTransferService' on port 34349.
26/06/02 11:15:01 INFO NettyBlockTransferService: Server created on 7362f7467f92:34349
26/06/02 11:15:01 INFO BlockManager: Using org.apache.spark.storage.RandomBlockReplicationPolicy for block replication policy
26/06/02 11:15:01 INFO BlockManagerMaster: Registering block manager 7362f7467f92:34349 with 366.3 MiB RAM, BlockManagerId(driver, 7362f7467f92, 34349, None)
26/06/02 11:15:01 INFO BlockManagerMaster: Registered block manager BlockManagerId(driver, 7362f7467f92, 34349, None)
26/06/02 11:15:01 INFO BlockManager: Initialized block manager: BlockManagerId(driver, 7362f7467f92, 34349, None)
26/06/02 11:15:02 INFO SharedState: Setting hive.metastore.warehouse.dir ('null') to the value of spark.sql.warehouse.dir.
26/06/02 11:15:02 INFO SharedState: Warehouse path is 'file:/spark-warehouse'.
=====
Part 1: Using Spark RDD
=====
District 1: Average Income = 163214.29
District 2: Average Income = 103600.00
District 3: Average Income = 59300.00
District 4: Average Income = 30571.43
District 5: Average Income = 20888.89
```

Spark SQL Output

```
what's next:
  Try Docker debug for seamless, persistent debugging tools in any container or image - docker debug spark-master
  Learn more at https://docs.docker.com/gv/debug-c11/
Using Spark's default log4j profile: org/apache/spark/log4j-defaults.properties
26/06/02 11:15:08 INFO SparkContext: Running Spark version 3.2.1
26/06/02 11:15:08 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
26/06/02 11:15:08 INFO ResourceUtils: =====
26/06/02 11:15:08 INFO ResourceUtils: No custom resources configured for spark.driver.
26/06/02 11:15:08 INFO ResourceUtils: =====
26/06/02 11:15:08 INFO SparkContext: Submitted application: Assignment2.AverageIncome_RDD
26/06/02 11:15:08 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, amount: 1, script: , vendor: , memory -> name: memory, amount: 1024, script: , vendor: , of
Heap -> name: offHeap, amount: 0, script: , vendor: ), task resources: Map(cpus -> name: cpus, amount: 1.0)
26/06/02 11:15:08 INFO ResourceProfile: Limiting resource is cpu
26/06/02 11:15:08 INFO ResourceProfileManager: Added ResourceProfile id: 0
26/06/02 11:15:08 INFO SecurityManager: Changing view acls to: root
26/06/02 11:15:08 INFO SecurityManager: Changing modify acls to: root
26/06/02 11:15:08 INFO SecurityManager: Changing view acls groups to:
26/06/02 11:15:08 INFO SecurityManager: Changing modify acls groups to:
26/06/02 11:15:08 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view permissions: Set(root); groups with view permissions: Set(); users with modify permissions:
Set(root); groups with modify permissions: Set()
26/06/02 11:15:09 INFO Utils: Successfully started service 'sparkDriver' on port 32823.
26/06/02 11:15:09 INFO SparkEnv: Registering MapOutputTracker
26/06/02 11:15:09 INFO SparkEnv: Registering BlockManagerMaster
26/06/02 11:15:09 INFO BlockManagerMasterEndpoint: Using org.apache.spark.storage.DefaultTopologyMapper for getting topology information
26/06/02 11:15:09 INFO BlockManagerMasterEndpoint: BlockManagerMasterEndpoint up
26/06/02 11:15:09 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
26/06/02 11:15:09 INFO BlockManager: Created local directory at /tmp/blockmgr-a2903e1-dbd1-4ec5-a233-94a6ce7e451c
26/06/02 11:15:09 INFO MemoryStore: MemoryStore started with capacity 366.3 MiB
26/06/02 11:15:09 INFO SparkEnv: Registering OutputCommitCoordinator
26/06/02 11:15:09 INFO Utils: Successfully started service 'sparkui' on port 4040.
26/06/02 11:15:09 INFO SparkUI: Bound SparkUI to 0.0.0.0, and started at http://7362f7467f92:4040
26/06/02 11:15:09 INFO Executor: Starting executor ID driver on host 7362f7467f92
26/06/02 11:15:09 INFO Utils: Successfully started service 'org.apache.spark.network.netty.NettyBlockTransferService' on port 37001.
26/06/02 11:15:09 INFO NettyBlockTransferService: Server created on 7362f7467f92:37001
26/06/02 11:15:09 INFO BlockManager: Using org.apache.spark.storage.RandomBlockReplicationPolicy for block replication policy
26/06/02 11:15:09 INFO BlockManagerMaster: Registering block manager BlockManagerId(driver, 7362f7467f92, 37001, None)
26/06/02 11:15:09 INFO BlockManagerMasterEndpoint: Registering block manager 7362f7467f92:37001 with 366.3 MiB RAM, BlockManagerId(driver, 7362f7467f92, 37001, None)
26/06/02 11:15:09 INFO BlockManagerMaster: Registered block manager BlockManagerId(driver, 7362f7467f92, 37001, None)
26/06/02 11:15:09 INFO BlockManager: Initialized block manager: BlockManagerId(driver, 7362f7467f92, 37001, None)
26/06/02 11:15:10 INFO SharedState: Setting hive.metastore.warehouse.dir ('null') to the value of spark.sql.warehouse.dir.
26/06/02 11:15:10 INFO SharedState: Warehouse path is 'file:/spark-warehouse'.
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Part 2: Using Spark SQL
=====
-----
|district_id|average_income|
-----+-----
|1|163214.28571428571|
|2|103600.0|
|3|59300.0|
|4|30571.428571428572|
|5|20888.888888888889|
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what's next:
  Try Docker debug for seamless, persistent debugging tools in any container or image - docker debug spark-master
  Learn more at https://docs.docker.com/gv/debug-c11/
bin\windows\MSYS_NO_PATHCONV=1 MINGW64 ~/OneDrive/Documents/gitHub/DAD66002/hw (main)
$
```